Advanced Computer Networks

Dr. Ahmad Khonsari

Fall 1405

Teaching Staff

- Instructor:
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- Chief Teaching assistant:
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Course Goals

- Learn principles of advanced computer networking and services
- Learn the basics of research in computer networking
- Hands-on implementation of advanced networking topics

Course Format

- ~16 lectures Slides: Review of Computer Networking course + BGP + SDN + NDN + Data Center + Wireless Networking + IPTV (IPv6, Cloud, IoT)
- Cover the "principles and practice"
- 2 Paper Reviews
- 3 homework assignments
  - ✓ All files and reports should be written in Latex
- 2 programming projects (Mainly implementations of the theoretical topics discussed in the course)
- Due to the rapid advancement in the research domain, some lectures might be altered, or added to the course materials on course webpage.

Grading

10% for Quiz/exams

40% for programming projects
20% for homework assignments
20% for video reviews
10% for paper reviews

Course Materials

- Course Slides on UT E-Learn
  - All materials are prepared to be delivered in a self-learning approach. Thus, students should carefully follow the lectures themselves.
  - If any part is not clear, first, go through the lecture again, then, use Internet , finally you may ask your questions from Instructor

Course Materials

- Textbook for further reading:

J. Kurose and K. Ross, Computer Networking –A Top-Down Approach, 8th ed. 2021 Pearson,

- L. L. Peterson and B. S. Davie,"Computer Networks, A Systems Approach", 6th ed. Morgan Kaufmann, 2021

. Andrew S. Tanenbaum, Nick Feamster, David J. Wetherall, Computer Networks. 6th ed. Pearson , 2021

https://sdn.systemsapproach.org/

Class rules

- Class attendance (In the case of announcement by authorities)
- **do not bring electronic devices , especially mobile phones in the class**
- **LATE ARRIVAL**: please do not enter the class after me

Class rules-project late

- see E-Learn page of the course: Advanced Computer Networks